TAR System Description Paper Template

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Abstract

This document provides the instructions on formatting the TAR system description paper in LaTeX. This is where you write the abstract (i.e., summary) of the work you carried out within the project. The abstract is a paragraph of text ranging between 70 and 150 words.

1. Introduction

This section is the introduction to your paper. Introduction should not be too elaborate, as that is what other sections are for (the Introduction should definitely not spill over to the second page).

This is the second paragraph of the introduction. In LaTeX, paragraphs are separated by inserting an empty line in between them. Avoid very large paragraphs (larger than half of the page height), but also avoid tiny paragraphs (e.g., one-sentence paragraphs).

2. Related work

In scientific papers, this section usually (but not necessarily) briefly describes the related research and what makes the presented approach different from it.

2.1. First subsection

This is a subsection of the second section.

2.2. Second subsection

This is the second subsection of the second section. Referencing the (sub)sections in text is performed as follows: "in Section 2.1. we have shown ...".

2.2.1. Sub-subsection example

This is a sub-subsection. If possible, it is better to avoid sub-subsections.

3. Methodology

In this section, we describe the established framework for evaluating the efficacy and robustness of pairing Negative Preference Optimisation (NPO) with Low-Rank Adaptation (LoRA) for large language model unlearning. We detail the datasets, models, and evaluation metrics employed to assess the performance of such an approach.

3.1. Tasks and dataset

The main dataset used in experiments is part of the *LUME: LLM Unlearning with Multitask Evaluations* benchmark. The dataset consists of documents that are categorised into three scenarios:

(I) **Synthetic creative documents** This scenario consists of synthetic fictional stories of different genres.

(II) **PII-containing synthetic biographies** This scenario consists of synthetic biographies of public figures that contain personally identifiable information (PII) such as names, home addresses, and Social Security numbers.

(III) **Real documents from pretraining corpus** This scenario consists of real documents that were part of the pretraining corpus of the language model.

Each of the subtasks of the challenge contains two main data splits: *Retain* and *Forget*. The *Retain* split contains documents that should be retained in the model's memory, while the *Forget* split contains documents that the model should unlearn. The number of examples can be found in Table 1.

Table 1: Distribution of documents in the dataset.

Document	Forget	Retain	Total
Synthetic Creative Documents	166	206	372
Synthetic Biographies	642	612	1254
Real Documents	304	318	622
Total	1112	1136	2248

3.2. Model

Our primary experiments are conducted using the *OLMo-7B-0724-Instruct-hf* model, which has been explicitly fine-tuned to memorise the documents from the LUME benchmark dataset.

Unlearning the above-mentioned model is achieved through Negative Preference Optimisation (NPO). As this method is computationally expensive, we apply it in combination with Low-Rank Adaptation (LoRA) to reduce the number of trainable parameters, and thus make the unlearning process more efficient. All experiments were conducted on a single NVIDIA A100 40GB GPU.

Given a prompt x, the training objective of our approach is to minimise the likelihood of generating forgotten documents y_f, relative to a fixed reference model π_{ref}. Formally, the training objective can be expressed as follows:

$$\mathcal{L}_{NPO}(\theta) = \frac{2}{\beta} \log \left(1 + \left(\frac{\pi_\theta(y_f|x)}{\pi_{ref}(y_f|x)} \right)^{\beta} \right),$$

where π_θ is the model being trained, π_{ref} is the reference model, and β is a hyperparameter that controls the strength of the preference. Furthermore, to preserve the model's general capabilities, we include an additional cross-entropy loss term that encourages the model to retain the knowledge of the retained documents y_r. The final loss function is:

$$\mathcal{L}(\theta) = \mathcal{L}_{NPO}(\theta) + \alpha \cdot \mathcal{L}_{CE}(\theta),$$

where $\mathcal{L}_{CE}(\theta)$ is the cross-entropy loss computed on the retained documents, and α is a hyperparameter that aids in balancing unlearning and knowledge retention.

As an inference-time baseline for jailbreaking comparison, we implemented Vector Steering (VS)...

3.3. Adversarial Red-Teaming

To evaluate the robustness of NPO-LoRA against adversarial attacks, we implemented an adversarial red-teaming procedure. We utilized *gpt-4o-mini* to rewrite the queries from the *Forget* split into seven different adversarial prompts:

- **Paraphrase** - altering the wording of the original query while preserving its meaning.
- **Prefix Injection** - prepends bypass instructions to the original query.
- **Roleplay** - frames the query as a roleplay scenario.
- **Translation** - translates the original query to another language.
- **Code Context** - embeds the original query within a code snippet.
- **Neighbour** - targets adjacent documents without referencing the original query.
- **Logical Chain** - multi-step reasoning that leads to the original query.

3.4. Evaluation Metrics

Our evaluation framework relies on the competition metrics, which aggregate three core dimensions via an arithmetic mean to generate a final submission score. First, task-specific regurgitation rates using ROUGE-L and Exact Match (EM) metrics are computed across the *Retain* and *Forget* splits, where the forget split metrics are inverted ($1-$ value). These 12 scores across the three tasks are then aggregated via the harmonic mean into a single score that reflects the overall performance of the unlearning approach. Second, a Membership Inference Attack (MIA) score is computed over member and non-member documents to assess the model's vulnerability to privacy attacks. Third, general capabilities are evaluated using the MMLU benchmark, and the performance is measured as the average accuracy across all tasks.

4. Figures and tables

4.1. Figures

Here is an example on how to include figures in the paper. Figures are included in LaTeX code immediately *after* the text in which these figures are referenced. Allow LaTeX to place the figure where it believes is best (usually on top of the page of at the position where you would not place the figure). Figures are referenced as follows: "Figure **??** shows ...". Use tilde (˜) to prevent separation between the word "Figure" and its enumeration.

4.2. Tables

There are two types of tables: narrow tables that fit into one column and a wide table that spreads over both columns.

4.2.1. Narrow tables

Table **??** is an example of a narrow table. Do not use vertical lines in tables – vertical tables have no effect and they make tables visually less attractive. We recommend using *booktabs* package for nicer tables.

4.3. Wide tables

Table 2 is an example of a wide table that spreads across both columns. The same can be done for wide figures that should spread across the whole width of the page.

5. Math expressions and formulas

Math expressions and formulas that appear within the sentence should be written inside the so-called *inline* math environment: $2+3$, $\sqrt{16}$, $h(x) = \mathbf{1}(\theta_1 x_1 + \theta_0 > 0)$. Larger expressions and formulas (e.g., equations) should be written in the so-called *displayed* math environment:

$$b_k^{(i)} = \begin{cases} 1 & \text{if } k = \text{argmin}_j \|\mathbf{x}^{(i)} - \mu_j\|, \\ 0 & \text{otherwise} \end{cases}$$

Math expressions which you reference in the text should be written inside the *equation* environment:

$$J = \sum_{i=1}^{N} \sum_{k=1}^{K} b_k^{(i)} \|\mathbf{x}^{(i)} - \mu_k\|^2 \tag{1}$$

Now you can reference equation (1). If the paragraph continues right after the formula

$$f(x) = x^2 + \varepsilon \tag{2}$$

like this one does, use the command *noindent* after the equation to remove the indentation of the row.

Multi-letter words in the math environment should be written inside the command *mathit*, otherwise LaTeX will insert spacing between the letters to denote the multiplication of values denoted by symbols. For example, compare $Consistent(h, \mathcal{D})$ and
$\mathit{Consistent}(h, \mathcal{D})$.

If you need a math symbol, but you don't know the corresponding LaTeX command that generates it, try *Detexify*.[1]

6. Referencing literature

References to other publications should be written in brackets with the last name of the first author and the year of publication, e.g., (Chomsky, 1973). Multiple references are written in sequence, one after another, separated by semicolon and without whitespaces in between, e.g., (Chomsky,

[1] `http://detexify.kirelabs.org/`

Table 2: Wide-table caption

Heading1	Heading2	Heading3
A	A very long text, longer that the width of a single column	128
B	A very long text, longer that the width of a single column	3123
C	A very long text, longer that the width of a single column	−32

1973; Chave, 1964; Feigl, 1958). References are typically written at the end of the sentence and necessarily before the sentence punctuation.

If the publication is authored by more than one author, only the name of the first author is written, after which abbreviation *et al.*, meaning *et alia*, i.e., and others is written as in (Johnson et al., 1976). If the publication is authored by only two authors, then the last names of both authors are written (Johnson and Howells, 1974).

If the name of the author is incorporated into the text of the sentence, it should not be in the brackets (only the year should be there). E.g., "Chomsky (1973) suggested that ...". The difference is whether you reference the publication or the author who wrote it.

The list of all literature references is given alphabetically at the end of the paper. The form of the reference depends on the type of the bibliographic unit: conference papers, (Chave, 1964), books (Butcher, 1981), journal articles (Howells, 1951), doctoral dissertations (Croft, 1978), and book chapters (Feigl, 1958).

All of this is automatically produced when using BibTeX. Insert all the BibTeX entries into the file `tar2023.bib`, and then reference them via their symbolic names.

7. Conclusion

Conclusion is the last enumerated section of the paper. It should not exceed half of a column and is typically split into 2–3 paragraphs. No new information should be presented in the conclusion; this section only summarizes and concludes the paper.

Acknowledgements

We would like to thank the University Computing Centre (SRCE) in Zagreb for providing the computational resources on their Advanced Computing platform used in this work.

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